Machine Intelligence Research
www.mi-research.net

22(5), October 2025, 983-998
DOI: 10.1007/s11633-023-1477-x

Motion-guided Visual Tracking

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Abstract: Motion information is a crucial cue to build a robust tracker, especially in handling object occlusion and fast drift caused by cameras and objects. However, it has not been fully exploited. In this study, we attempt to exploit motion cues to guide visual trackers without bells and whistles. First, we decouple motion into two types: camera motion and object motion. Then, we predict them individually via the proposed camera motion modeling and object trajectory prediction. Each module contains a motion detector and a verifier. As for camera motion, we apply the off-the-shelf keypoint matching method to detect camera movement and propose a novel self-supervised camera motion verifier to validate its confidence. Given the previous object trajectory, object trajectory prediction aims to predict the future location of the target and select a reliable trajectory to handle fast object motion and occlusion. Numerous experiments on several mainstream tracking datasets, including OTB100, DTB70, TC128, UAV123, VOT2018 and GOT10k, demonstrate the effectiveness and generalization ability of our module, with real-time speed.

Keywords: Visual tracking, motion embedding, camera motion modeling, self-supervised learning, trajectory prediction.

Citation: P. Zhang, S. Lai, D. Wang, H. Lu. Motion-guided visual tracking. *Machine Intelligence Research*, vol.22, no.5, pp.983–998, 2025. http://doi.org/10.1007/s11633-023-1477-x

1 Introduction

Given the target position in the initial frame, visual object tracking aims to estimate the target location during the whole sequence. In the early stage, several tracking frameworks, including Kalman filter[1, 2] and particle filter[3, 4], utilize motion information to track the target. These approaches locate the target position via the target′s previous trajectory and velocity. Recently, with the emergence of powerful hand-crafted and deep features, appearance-based trackers have achieved significant accuracy. These trackers search for the target candidate with the maximum response within a predefined search region. Several popular appearance-based frameworks have been presented, including correlation filters[5, 6], Siamese networks[7, 8], and multi-domain networks[9, 10], all of which exhibit excellent tracking performance and fast speed. While appearance is often more reliable and meaningful than motion cues, its performance can be limited when the appearance information for tracking is unreliable, as described in Fig. 1. On the other hand, motion cues are insensitive to dramatic scene variations, making them an important supplement to the tracking of appearance information.

Based on the aforementioned observations, we propose an effective module to embed motion cues to assist appearance-based trackers, which can be generalized to most recent tracking frameworks. Our contributions can be summarized as follows:

1) We propose a camera motion modeling module, which consists of a camera motion detector and a camera motion verifier. We apply an off-the-shelf keypoint matching method to detect camera motion and propose a novel camera motion verifier via a self-supervised framework to obtain the confidence of camera motion.

2) We design an object trajectory prediction module, including an object trajectory detector and an object trajectory verifier. As for the object trajectory detector, we utilize the Conv-LSTM to predict the target′s future location and velocity by using the previous motion information. Then, the predicted results are sent to a classification network to evaluate its confidence.

3) We formulate both camera and object motions in a unified framework to achieve a powerful supplement to appearance-based trackers without bells and whistles.

4) Comparison results on six datasets, including OTB100, DTB70, TC128, UAV123, VOT2018 and GOT10k, validate the effectiveness of our motion module. Notably, our method can be equipped with most of the latest tracking frameworks to further increase their accuracy with negligible cost.

2 Related works

2.1 Visual object tracking

In the last decade, visual object tracking has been de-

Research Article
Manuscript received on June 26, 2023; accepted on September 14, 2023; published online on August 19, 2025
Recommended by Associate Editor Jordi Gonzalez
Colored figures are available in the online version at https://link.springer.com/journal/11633
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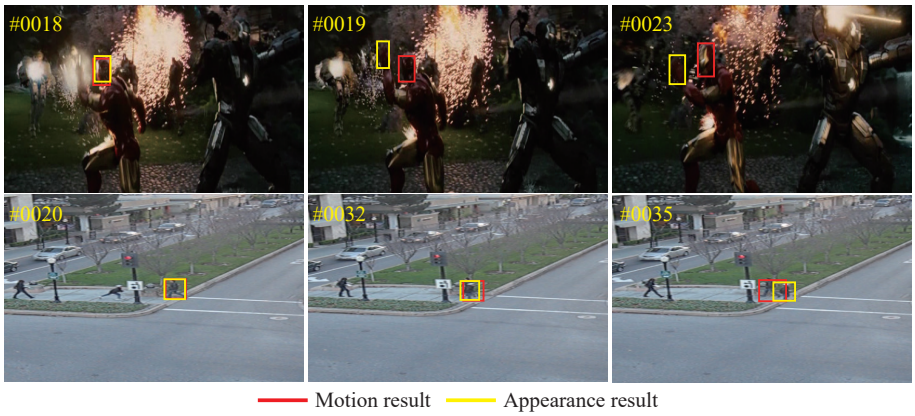


Fig. 1 Comparison between motion-based and appearance-based trackers. Taking DIMP[11] and our motion-guided module as appearance and motion trackers, we find that motion cue has superior performance in handling occlusion, dramatic movement and tiny object capturing, which acts as a supplement to the appearance tracker. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

veloped rapidly via both motion and appearance cues. According to the adopted information, tracking methods can be categorized into motion-based trackers (MT) and appearance-based trackers (AT). MT methods aim to predict the target location via motion information, including the previous trajectory velocity and optical flow. The typical MT frameworks are based on Kalman filter[1, 12–14], particle filter[4, 15–17], and optical flow[18–20]. Motion information is insensitive to target variation and often achieves robust performance in challenging scenes, such as illumination variation, target occlusion and background clustering. However, depending on previous target's state, motion cues are not always reliable due to the arbitrariness of target motion.

On the other hand, appearance-based trackers learn a target model, which is used for localization via template matching or tracking-by-detection approach. The performance mainly depends on the discriminability of features used for target modeling. With the emergence of powerful hand-crafted and deep features, AT has dominated this field with high performance and speed. Several frameworks, such as correlation filter[5, 6, 21–23], Siamese network[7, 8, 24], and multi-domain network[9, 10] have been proposed, whose performance are significantly superior to MT. Compared with MT, AT suffers from target draft caused by dramatic appearance variation. Though the online updating strategy learns target variation during tracking, trackers cannot perform well under extreme conditions.

Recently, motion-assisted appearance trackers have attracted increasing attention. Li and Yeung[25] predict camera motion to handle tracking with drones. Liu et al.[26] aim to model background motion and target trajectory using a spatial-temporal network. Those trackers combine both appearance and motion cues, which can be used to build a more robust model. In this work, we formulate both camera and object motions in a unified framework, to further improve the tracking accuracy of the AT algorithms.

2.2 Contrastive self-supervised learning

Contrastive self-supervised learning (CSL), as a mainstream of self-supervised learning, was first proposed by Hadsell et al.[27] and has been widely applied in many computer vision tasks[28–31]. The CSL algorithms attempt to learn the feature representation by optimizing a pretext task, which can provide supervised label information to exploit the potential characteristic of numerous unlabeled data and contrast triplet training pairs that contain anchor, positive and negative samples. Several pretext tasks, including image colorization[32], in-painting[33], and image transformation[34] are adopted. Recently, researchers have focused on designing loss functions[35, 36] and similarity metrics[30, 33]. Mnih and Kavukcuoglu[35] design a novel negative contrastive estimation loss for CSL. A simple framework for CSL[37] is proposed without requiring specialized architectures or a memory bank. Schroff et al.[36] utilize CSL for facial recognition via triplet loss, thereby obtaining state-of-the-art performance among supervised competitors. In this work, we adopt the idea of CSL to study the camera motion problem, since this problem naturally lacks supervised information.

2.3 Object trajectory prediction

Object trajectory prediction is a sequence-based prediction task, that predicts the location of interest via past observations. Traditionally, object trajectory prediction focuses on the target in a specific category, including pedestrian[38–40], vehicle[41–43], to name a few. Although the target can move arbitrarily, the object's behavior can be driven by its motivation, scene condition, and interaction with other objects. According to these available priors, object trajectory prediction methods can be mainly categorized into three types: interaction-aware[38, 39, 42],

scene-aware[40, 43], and hybrid[41] methods. Zhang et al.[38] propose state-refinement LSTM to model the interaction between the object and its neighbors. Liang et al.[40] encode scene data into a grid map and adopt ConvRNN for location prediction. Tang and Salakhutdinov[41] introduce a probabilistic framework to learn latent variables using both scene and interaction information. Motivated by these discussions, we present a novel method to predict the object trajectory and verify the prediction result, to construct an effective motion-guided module.

3 Proposed approach

In this section, we detail our approach in Fig. 2. First, we decouple the overall offset into two aspects: 1) camera motion offset and 2) object motion offset. To estimate those offsets effectively, our method consists of two main components: camera motion modeling and object trajectory prediction. Each module contains a motion detector and verifier. The motion detector aims to estimate the corresponding offset, and then the verifier evaluates the confidence of detectors to filter out unreliable motion features. After that, the offset predicted by the motion cue is embedded to provide a reliable search region for the appearance model to capture the target. Compared with the popular tracking algorithms, our proposed motion-guided visual tracker fully exploits motion cues to assist the appearance model, especially in severe motion and occlusion cases.

3.1 Camera motion modeling (CMM)

Camera motion, accompanied by large object offset and motion blur, often occurs when a video is captured by handheld and wearable devices and becomes a well-known challenge in visual tracking. Existing appearance tracking paradigms solve this issue in the brute-force method, which searches the candidate using discriminative features and enlarges the search region when the target is not found. Although few efforts have been made, the problem has not been solved fundamentally. This section aims to obtain accurate camera movement via a camera motion detector and camera motion verifier.

3.1.1 Camera motion detector (CMD)

The CMD module aims to model the camera pose variation that can be formulated as $\langle \mathbf{R}, \mathbf{t} \rangle$. $\mathbf{R} \in \mathrm{SO}(3)$ is a rotation matrix and $\mathbf{t} \in \mathbb{R}^{1\times 3}$ is the translation vector representing the translation in the 3D Cartesian coordinate system. However, in RGB tracking, camera parameter and depth information are not provided for precise camera pose estimation. One suboptimal solution is to estimate the camera movement in image plane by calculating the transformation matrix between adjacent frames[25, 44]. Following this idea, we apply off-the-shelf keypoint extraction and matching to obtain the transformation matrix. First, the keypoints and descriptors of the adjacent frames $\mathbf{I}_{t-1}, \mathbf{I}_t \in \mathbb{R}^{W\times H\times 3}$ are extracted using existing methods, such as scale-invariant feature transform (SIFT)[45], speeded-up robust features (SURF)[46], and the latest deep-based features with GPU acceleration[47, 48]. The symbols W and H denote the image width and height, respectively. Then, the keypoints are matched according to their distance of descriptors. The transformation matrix[1] $\mathbf{M} \in \mathbb{R}^{2\times 3}$ can be obtained by solving (1).

$$\mathbf{M} = \mathcal{M}(\mathbf{I}_{t-1}, \mathbf{I}_t). \tag{1}$$

The function $\mathcal{M}(\cdot,\cdot)$ denotes the keypoint extraction and matching operation between two images. Then, the offset $\mathbf{O}_{CM}$ caused by camera motion is defined as

$$\begin{aligned} \mathbf{c}' &= \mathbf{M} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix} \\ \mathbf{O}_{CM} &= \mathbf{c}' - \mathbf{c} \end{aligned} \tag{2}$$

where $\mathbf{c} \in \mathbb{R}^{2\times 1}$ denotes the coordinate of target center in $\mathbf{I}_{t-1}$ and $\mathbf{c}'$ is the warped coordinate.

3.1.2 Camera motion verifier (CMV)

During tracking, complex scene variations, such as illumination variation and motion blur, may lead to misalignment of keypoints, and the accumulative error caused by camera motion cannot be omitted, which leads to tracking drift. To this end, we propose a CMV network to evaluate the performance of CMD and output a confidence score S_{CM}.

Inference. As shown in Fig. 3, taking the warped image $\mathbf{I}'_{t-1}$ (transformed from $\mathbf{I}_{t-1}$ using matrix $\mathbf{M}$) and current image $\mathbf{I}_t$ with their keypoint map $\mathbf{K}$ as input, the CMV with the Siamese architecture consists of 5 convolutional (Conv) layers for feature extraction and 3 fully-connected (FC) layers for confidence regression, which is followed by a sigmoid function. Note that the keypoint map $\mathbf{K} \in \mathbb{R}^{W\times H\times 1}$ aims to encode the extracted keypoints, which can be expressed in (3). The detailed implementation of the CMV can be found in the supplementary file.

$$\mathbf{K}(x,y) = \begin{cases} 1, & \text{if } (x,y) \text{ is a keypoint} \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Training. Learning CMV needs to overcome two realistic difficulties: 1) The ground truth of camera pose variation is not provided in tracking datasets; and 2) the camera motion is often accompanied by target movement and interference, including motion blur, illumination variation, to name a few, affects the evaluation performance.

To solve these difficulties, we generate much training

[1] Our method takes affine transformation as an example and can be applied to all other camera models.

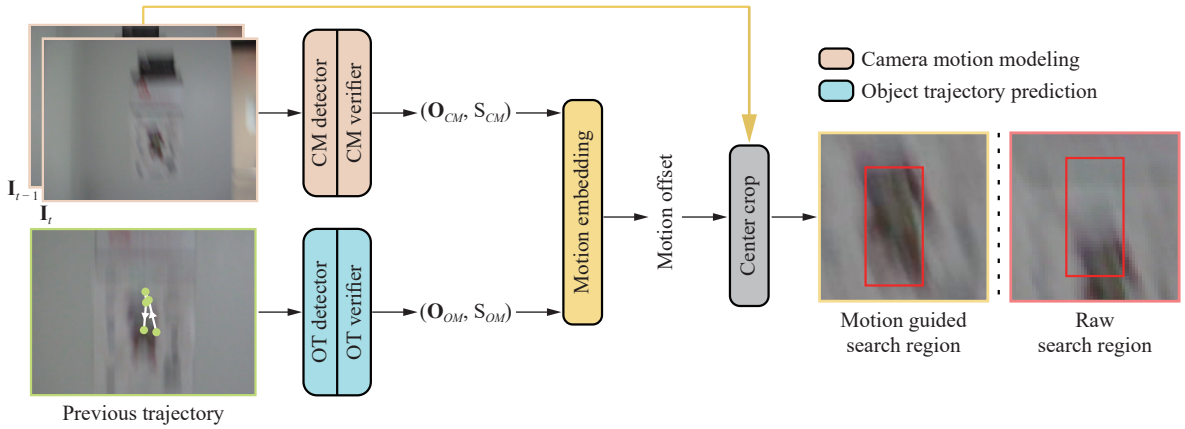


Fig. 2 Overall workflow of the proposed motion-guided trackers (Colored figures are available in the online version at https://link.springer.com/journal/11633)

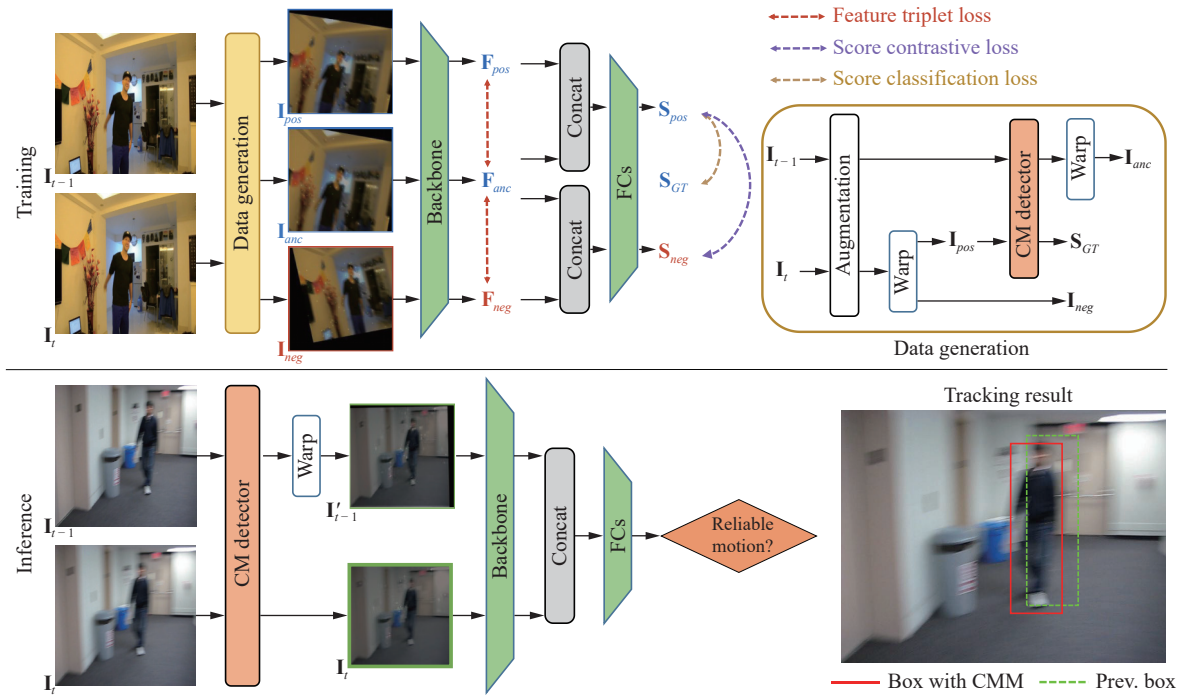


Fig. 3 Workflow of the proposed CMV. The CMV employs a contrastive self-supervised learning framework to exploit the potential features using unlabelled data. Our CMV is capable of measuring the confidence of the camera motion output and is robust under extreme conditions such as motion blur and severe illumination. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

data to simulate camera movement and other challenges and employ contrastive self-supervised learning to exploit useful information using unsupervised data.

Data generation. Following the pipeline of CSL, the data generation process aims to generate anchor, positive and negative samples used for contrastive representation learning. To simulate tracking scenes, we select adjacent frames to construct training pairs. The positive samples are obtained by $\mathbf{I}_{pos} = \mathcal{T}(\mathcal{A}(\mathbf{I}_t); \mathbf{M}_{pos})$ ($\mathcal{A}$ is the stochastic data augmentation operation), which applies brightness adjustment, motion blur and Gaussian blur to the given image to eliminate the influence of difficulty 2. The function $\mathcal{T}(\cdot; \mathbf{M})$ denotes the image transformation operation using matrix $\mathbf{M}$. $\mathbf{M}_{pos}$ is randomly generated. Similar to positive samples, negative samples are generated by $\mathbf{I}_{neg} = \mathcal{T}(\mathcal{A}(\mathbf{I}_t); \mathbf{M}_{neg})$. Finally, the anchor image is obtained by $\mathbf{I}_{anc} = \mathcal{T}(\mathcal{A}(\mathbf{I}_{t-1}); \mathbf{M}_{res})$ where $\mathbf{M}_{res}$ is obtained by matching $\mathbf{I}_{t-1}$ and $\mathbf{I}_{pos}$ via the camera motion detector, which can be expressed as $\mathbf{M}_{res} = \mathcal{M}(\mathbf{I}_{t-1}, \mathbf{I}_{pos})$. During training, we evaluate the confidence of camera motion by measuring the distance between the output transformation matrix $\mathbf{M}_{res}$ and the ground truth $\mathbf{M}_{pos}$ via L1-norm. Finally, the ground truth of confidence, S_{GT} is defined as

$$S_{GT} = \begin{cases} 1, & \text{if } ||\mathbf{M}_{res} - \mathbf{M}_{pos}||_1 < t_{CM} \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Loss function. We propose a loss function at the score and feature levels, which contains score classification loss, score contrastive loss and feature triplet loss.

1) Score classification loss: This loss aims to minimize the distance between the predicted score S_{CM} and label S_{GT}, which is defined as

$$\mathcal{L}_{cls} = ||S_{CM} - S_{GT}||^2. \quad (5)$$

2) Score contrastive loss: Followed by CSL, we form a score contrastive loss to enlarge the distance between positive and negative samples.

$$\mathcal{L}_{con} = \max(S_{CM}^{neg} - S_{CM}^{pos} + C_{con}, 0). \quad (6)$$

3) Feature triplet loss: The image distance is often related to the performance of the CMD. Thus, we employ a hard-negative mining strategy to obtain a better embedding space for distinguishing whether CMD is reliable. Specifically, we adopt a triplet loss concerning the features as

$$\mathcal{L}_{tri} = \max(||\mathbf{F}_{anc} - \mathbf{F}_{pos}||_2 - ||\mathbf{F}_{anc} - \mathbf{F}_{neg}||_2 + C_{tri}, 0). \quad (7)$$

Finally, the overall loss is expressed as

$$\mathcal{L}_{CMV} = \mathcal{L}_{cls} + \mathcal{L}_{con} + \mathcal{L}_{tri}. \quad (8)$$

3.2 Object trajectory prediction (OTP)

When the target is heavily occluded or in low resolution, the discriminability of appearance-based trackers is degraded, since the target appearance is not reliable for accurate tracking. Furthermore, when the target moves fast, trackers may fail to locate the target with the limited search region. Based on this observation, we aim to exploit object motion information to assist tracking via the proposed OTP module, which consists of an object trajectory detector (OTD) and an object trajectory verifier (OTV). OTD aims to predict the target's location and velocity in the following frames with the guidance of the previous trajectory. Then, OTV aims to evaluate whether the prediction is a target or not and outputs its confidence. Different from the trajectory prediction network in [26], which produces the future location of the target using both target previous trajectory and scene information, we model this problem solely depending on the motion cue and achieve a scene-independent motion predictor, thereby building a more general motion tracker.

3.2.1 Object trajectory detector (OTD)

As shown in Fig. 4, given the motion vectors $\mathbf{x}^t = [c_x^t, c_y^t, v_x^t, v_y^t]$ $(t \in [T-s, \cdots, T-1])$ in previous s frames, OTD aims to learn temporal dependency by the recurrent neural network, where (c_x, c_y) denotes the coordinate of the target center and (v_x, v_y) represents the target velocity. The initial position and various image sizes lead to a strong bias in learning specific scenes. To avoid this issue, we first normalize $\mathbf{x}$ and embed it via a position encoder, providing a fixed-size motion map $\mathbf{G}^t \in \mathbb{R}^{g \times g}$ based on the motion cue $(\mathbf{G}^t \sim \mathcal{N}(\frac{g \times c_x^t}{W}, \frac{g \times c_y^t}{H}, \frac{g \times v_x^t}{W}, \frac{g \times v_y^t}{H}, 0))$. The predicted motion vector is obtained by

$$\mathbf{x}^{t+1} = d(h(\text{LSTM}(f(\mathbf{G}^t)))) \quad (9)$$

where $f(\cdot)$ denotes two Conv layers whose kernel size is

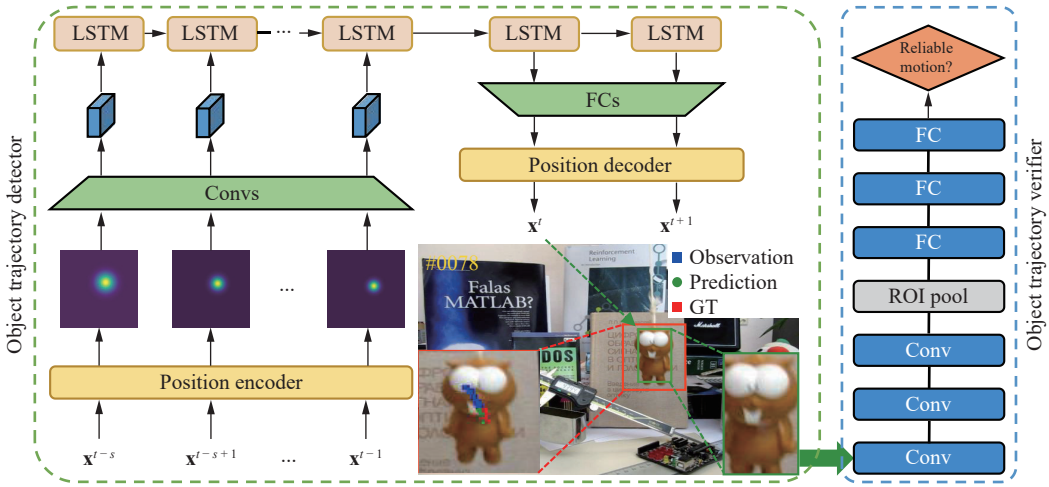


Fig. 4 Architecture of object trajectory prediction module, which contains an OTD and OTV. The OTD aims to predict the future location via Conv-LSTM to handle temporal information and the OTV submodule evaluates the results of OTD. Our OTP makes a self-aware trajectory predictor without extra appearance information. (Colored figure is available in the online version at https://link.springer.com/journal/11633)

5×5, and $h(\cdot)$ represents two FC layers for regressing the Conv-LSTM's outputs to the normalized motion vector. Finally, the motion vector is mapped to the raw image size via the position decoder $d(\cdot)$.

The network is optimized by minimizing the L2-norm between the prediction motion vector and ground truth as

$$\mathcal{L} = \sum_{t=1}^{T_{pred}} ||\mathbf{x}^t - \mathbf{x}_{GT}^t||^2. \tag{10}$$

Then, the offset caused by object motion $\mathbf{O}_{OM}$ is obtained by calculating the distance between the prediction and the last observation.

$$\mathbf{O}_{OM} = [c_x^{\mathrm{T}} - c_x^{\mathrm{T}-1}, c_y^{\mathrm{T}} - c_y^{\mathrm{T}-1}]^{\mathrm{T}}. \tag{11}$$

3.2.2 Object trajectory verifier (OTV)

The tracked object moves arbitrarily in the tracking task, which increases the prediction difficulty. The tracking error will be accumulated when OTD gives biased results. We aim to filter out the unreliable predicted location via the proposed OTV module. We model this process as a classification task, which aims to distinguish whether the predicted candidate is the object in the first frame or not. The architecture of OTV is shown in Fig. 4, which consists of 3 Conv layers for feature extraction. To adapt the target to various sizes, we introduce a precise region of interest (ROI) pooling layer[49] to embed a fixed-size representation. Then, three FC layers are utilized to predict the similarity between the prediction and initial target. In the offline training process, the OTV aims to learn a general representation. We draw positive and negative samples from the frame in the same sequence, whose interaction over union (IoU) are larger than 0.7 and lower than 0.3, respectively. The network is optimized via minimizing the binary classification loss. During tracking, with the guidance of target appearance in the first frame, the FC layers are further fine-tuned to embed target-specific representation and fixed in the whole tracking process. In the following frames, the predicted target obtained by OTD is sent to the overall network to generate the confidence S_{OM}.

3.3 Motion embedding

The proposed motion module aims to correct the search region and provide tracking results when the appearance model is not reliable. When motion occurs, our camera and object motion modules decouple the motion and work individually. In the tracking step, the camera motion is first detected and then the offset caused by object motion is predicted by the object motion module. After that, we calculate the overall offset $\mathbf{O}$ via (12) and use it to correct the search region.

$$\mathbf{O} = r(S_{CM}) \times \mathbf{O}_{CM} + r(S_{OM}) \times \mathbf{O}_{OM} \tag{12}$$

where $r(\cdot)$ is the step function defined as

$$r(x) = \begin{cases} 1, & \text{if } x > t \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

The threshold t controls the reliability of each motion type. In object motion module, we record the object trajectory to eliminate the effect of camera movement.

Furthermore, to make a powerful supplement to appearance cues, motion information is used for tracking when the appearance tracker is not reliable. We follow the setting in discriminative model prediction (DIMP)[11]. If the appearance tracker is confused with the target location, we resort to motion to obtain the final result.

3.4 Implementation details

As for CMD, we employ SURF with GPU acceleration as our feature extractor and model the camera motion via affine transformation. For training CMV, we select large-scale single object tracking benchmark (LASOT)[50] as our training data. Since the camera motion in the raw dataset cannot be measured, which leads to biased learning, we select 122 sequences with still cameras for training. The training pairs are first resized to a fixed-size patch with a size of 300×300. Our network is optimized via stochastic gradient descent (SGD) with a total of 32 pairs per mini-batch. The learning rate and weight decay are 0.000 1 and 0.000 5, respectively. The trade-off weights for contrastive loss and triplet loss are all set to 0.5. The margins C_{con} and C_{tri} in CMV loss are 0.2 and 1.0. The threshold t_{CM} for S_{GT} is set to 60. During tracking, camera motion and object motion, whose confidence S_{CM} and S_{OM} are larger than 0.8 and 0.6, are adopted to correct the search region.

We utilize the same training set, 122 still sequences in LASOT, for training OTP, to reduce the effects caused by camera motion. The network is optimized with Adam[51] and the learning rate is set to 0.000 1. The size of motion map g is set to 63. We utilize the previous 5 frames as object trajectory and predict 3 steps of the motion vector in the future. As for fine-tuning the OTV network, we draw 500 positive and 5 000 negative samples following the metric described in Section 3.2.2 in the initial frame of each sequence. The OTV is trained with SGD with learning rate of 0.000 1. The fine-tuning process contains 50 epochs and is efficient with neglectable cost in tracking speed.

Our method is implemented by the PyTorch deep learning library, on a PC machine with an Intel-i9 9900k CPU (64 G RAM) and an RTX Titan GPU (24 G

memory). Both training and test codes will be publicly available.

4 Experiments

4.1 Dataset and metrics

OTB100[52]. OTB100 is a popular benchmark, which contains 100 video sequences annotated with 11 attributes. Two evaluation metrics including success rate (SR) and precision rate (PR) are used for analysis. SR calculates the percentage of frames whose IoU is larger than a threshold. PR measures the percentage of success frames where the distance between the bounding box and ground truth is lower than a threshold.

DTB70[25]. In recent years, tracking with drones has received increasing attention. To this end, DTB70 consists of 70 videos captured by drone cameras. In this dataset, large movements often occur caused by camera motion, which requires the tracker to handle large viewpoint variation well. SR and PR are used to evaluate the trackers.

TC128[53]. TC128 collects 128 color sequences to exploit the importance of color information for tracking. TC128 labels all the sequences with 11 attributes, which is the same as OTB100 and adopts the same evaluation metrics to measure trackers′ performance.

UAV123[54]. UAV123 dataset contains 123 video clips, which are captured by unmanned aerial vehicles (UAVs). The camera often moves dramatically, requiring motion compensation to correct the camera movement. We evaluate the trackers via SR and PR metrics.

VOT2018[55]. VOT2018 has 60 short-term video sequences, containing nearly 21 k frames. Different from other datasets, VOT evaluates the tracker in terms of accuracy (A) and robustness (R), which focus on IoU and failure time, respectively. The overall expected average overlap (EAO) combines both metrics in a unified protocol. More descriptions of VOT2018 can be referred to [56].

GOT10k[57]. GOT10k is a large-scale dataset containing over 10 k video clips, 180 of which are used for testing. Two metrics are used for evaluation including average overlap (AO) and the popular success rate (SR) with different thresholds (i.e., 0.5 and 0.75). To guarantee fair evaluation, we use the appearance tracker which is trained only on the internal training data.

Motion dataset 200 (MD200). The above datasets focus on constructing comprehensive scenes to evaluate the overall tracker performance. However, in real applications, dramatic motion handling is critical for tracker construction. To highlight the importance of motion information and evaluate trackers′ motion handling capability, we collect 200 videos from public datasets (74 videos are from ALOV300[58], 22 videos are from NfS[59] and 104 videos belong to NUS-PRO[60]), with large camera and object motion[2] and form a new dataset, namely MD200. The basic statistics of the MD200 are depicted in Table 1. We apply SR and PR to evaluate tested trackers on MD200. Fig. 5 shows some representative examples in MD200. More details can be found in the supplementary material.

Table 1 Basic statistics of the MD200 dataset

Datasets	Num. seq	Min. frame	Max. frame	Avg. frame	Med. frame
OTB[52]	100	71	3 872	590	393
TC128[53]	200	71	3 872	429	365
DTB70[25]	70	68	699	225	202
VOT2018[55]	60	41	1 500	356	302
MD200	200	22	2 063	287	250

4.2 Comparison with state-of-the-art trackers

Overall performance on OTB100. As shown in Table 2, compared with all competitors, DIMP with our motion-guided module, MG-DIMP, achieves the top performance with respect to the success rate. Furthermore, DIMP obtains 2.2% and 2.6% promotion in SR and PR, respectively.

Overall performance on DTB70. We conduct analysis on this dataset captured by unmanned aerial vehicle (UAV), in which large camera motion and viewpoint change need to be handled. MG-DIMP obtains the best results and performs better than the baseline tracker, achieving SR and PR scores of 64.9% and 84.9%, respectively.

Overall performance on TC128. Our tracker also achieves satisfactory performance on TC128, which focuses on encoding color information for tracking. Table 2 shows the SR and PR scores for 128 sequences in the dataset. MG-DIMP is superior to all the previous methods in terms of both SR and PR. Compared to the baseline tracker, our MG-DIMP achieves 1.8% and 2.1% improvements in SR and PR, respectively, which demonstrates the effectiveness of our proposed module.

Overall performance on UAV123. The proposed motion-guided tracker shows potential in handling the UAV-captured dataset, where large viewpoint change occurs frequently. As shown in Table 2, MG-DIMP is superior to the baseline, DIMP, with 1.7% promotion in both SR and PR evaluations.

Overall performance on VOT2018. We evaluate our approach on VOT2018 and report the performance in Table 3. MG-DIMP, which achieves a very competitive performance, performs better than the baseline with 3.0%

[2] The targets in the selected videos move dramatically, which is caused by either camera motion or object motion or both.

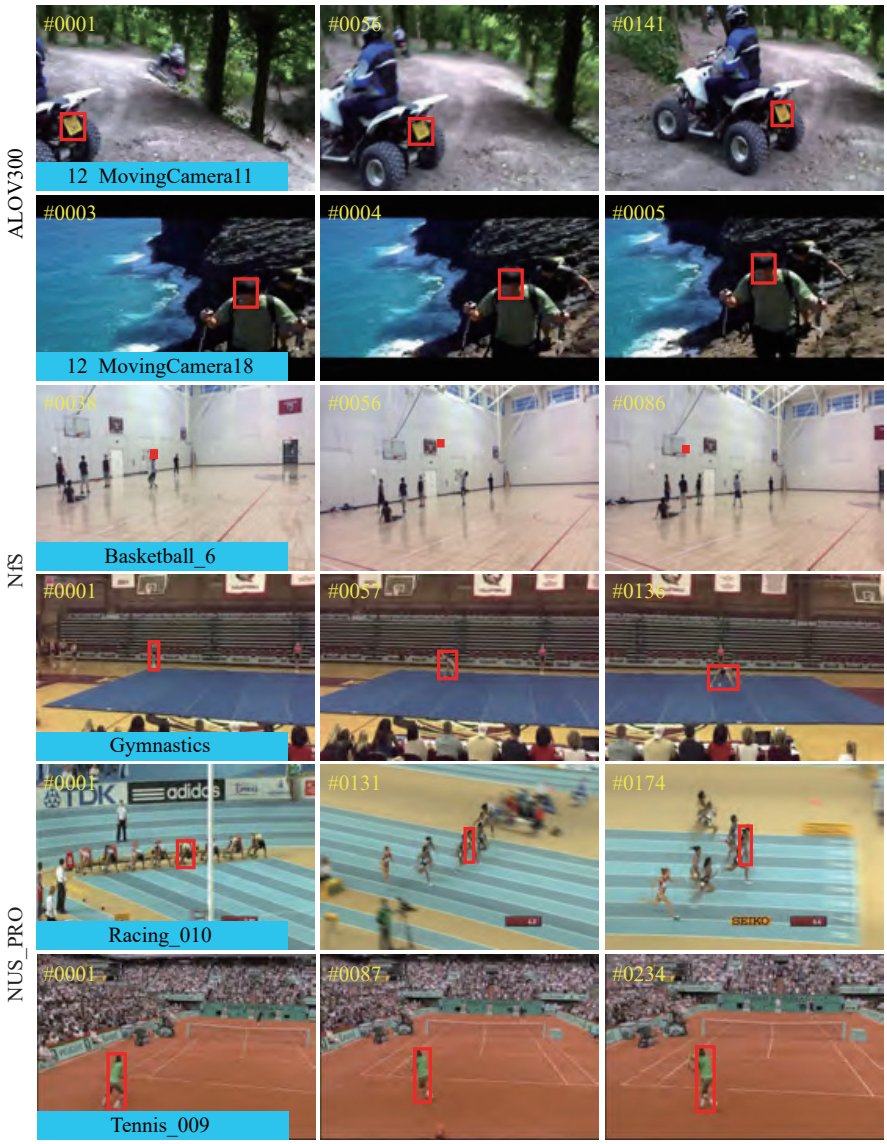


Fig. 5 Examples of our constructed motion dataset 200 (MD200) (Colored figures are available in the online version at https://link.springer.com/journal/11633)

Table 2 The success and precision rate of state-of-the-art trackers on several mainstream and our constructed datasets

	OTB100 SR/PR (%)	**DTB70** SR/PR (%)	**TC128** SR/PR (%)	**UAV123** SR/PR (%)	**GOT10k** AO/$SR_{0.5}$/$SR_{0.75}$ (%)	**MD200** SR/PR (%)
ECO[6]	66.4/90.4	55.3/81.9	58.2/80.8	53.6/76.9	39.5/40.7/17.0	55.6/65.4
RTMDNet[10]	61.7/84.5	45.2/69.3	57.2/78.8	48.6/72.1	40.4/42.4/14.7	50.1/57.0
SiamRPN[24]	66.6/87.6	57.2/79.2	55.5/75.7	57.6/76.7	44.0/52.2/20.7	52.7/60.8
$LADCF_{HC}$ [61]	65.9/85.5	42.5/61.9	55.1/73.9	49.4/70.2	28.0/27.6/10.3	51.5/60.1
SiamRPN++ [62]	69.5/**90.5**	59.1/79.3	57.6/77.6	61.1/80.4	51.8/61.8/32.5	53.7/62.7
DIMP[11]	68.1/88.0	64.1/83.6	60.6/80.2	65.0/86.0	59.5/70.8/46.2	58.8/68.5
MG-DIMP	**69.6**/90.3	**64.9/84.9**	**61.7/81.9**	**66.1/87.5**	**60.6/72.5/46.9**	**62.5/73.6**

promotion and 3.9% decrease in both EAO and robustness, respectively. The results show that motion cues can be effective supplements to appearance trackers and our motion-guided module can achieve more robust tracking results.

Overall performance on GOT10k. We evaluate

Table 3 The expected average overlap, accuracy and robustness of state-of-the-art trackers on VOT2018. Top performance is labeled in **red** font.

	EAO(↑)	A(↑)	R(↓)
ECO[6]	0.280	0.484	0.276
SiamRPN[24]	0.383	0.586	0.276
LADCF[61]	0.389	0.503	**0.159**
SiamRPN++ [62]	0.414	0.600	0.234
DIMP[11]	0.439	0.615	0.180
PG-Net[63]	0.447	**0.618**	0.192
MG-DIMP	**0.452**	0.610	0.173

our method on a large-scale dataset, GOT10k and the performance is listed in Table 2. Compared with all the competitors, our method achieves the highest score among those three metrics. In specific, MG-DIMP achieves a 1.8% and 2.4% improvement over the baseline (DIMP), which demonstrates that our motion-guided module can provide a more accurate target position and precise scale estimation.

Overall performance on MD200. In our constructed MD200 dataset, which aims to emphasize the importance of motion cues, all the competitors based on appearance model result in a significant decrease in tracking performance. Thus, the necessity to fully exploit motion information should be paid more attention. Our MG-DIMP obtains impressive results of 62.5% and 73.6% in SR and PR, which outperforms the baseline tracker by a large margin. The results show that our motion-guided module, modeling both camera and object motion as supplement cues to the appearance model, is capable of handling dramatic motion and appearance variation.

Attribute-based performance on OTB100. As shown in Table 4, we compare attribute-based performance with and without the motion-guided module on DIMP. Two attributes, including motion blur (MB) and fast motion (FM), are related to camera motion. MG-DIMP achieves 3.9% and 3.1% promotion according to SR in MB and FM, respectively. The performance improvement mainly stems from the effectiveness of our CMM module. Our OTP module is capable of providing reliable guidance when the target appearance is not reliable for tracking, especially in occlusion (OCC), low resolution (LR). The tracker with our motion-guided module performs better than the baseline in those two attributes with a large margin. The improvements on LR and OCC are 18.7% and 2.8% in terms of SR, respectively. All the results demonstrate the effectiveness of our model.

4.3 Ablation study

Analysis on different components. In this section, we detail the effectiveness of each component in MGVT. Taking DIMP as our appearance tracker, we compare several variants of MG-DIMP, including 1) DIMP: The baseline method without motion information; 2) DIMP+CMD: The baseline tracker with the CMD submodule described in Section 3.1.1; 3) DIMP+CMD+CMV: The camera motion is estimated by CMD and then selected by CMV; 4) DIMP+CMM+OTD: The baseline tracker is equipped with CMM, consisting of CMD and CMV. Furthermore, the object motion is estimated by the OTD network; 5) DIMP+CMM+OTD+OTV: The final method, where both camera motion and object motion are predicted and measured using motion detectors and verifiers. Then, the motion results are embedded to provide a reliable search region for the appearance tracker. The results on OTB100 and MD200 are shown in Table 5. Compared with the baseline tracker, DIMP+CMD+CMV achieves 1.8% and 2.2% promotion in OTB in terms of SR and PR, respectively. OTP module further contributes to final results in OTB and MD200, respectively. The results demonstrate that motion exploitation has great potential in building an accurate tracker, which can be applied to real scenes. However, trackers solely used motion detectors achieve decreasing results. The reason is that the complex tracking scenes, such as background hollowness and aimless movement of the target, significantly influence the performance of our motion detectors. The verifiers aim to evaluate the results of these detectors and filter out unreliable information, thereby leading to better results.

Table 4 Attribute-based comparison on OTB100 with baseline trackers. MG-DIMP achieves superior performance among all the attributes, especially in occlusion, fast motion, out of view, background clutter and low resolution.

	DIMP[11] SR/PR (%)	MG-DIMP SR/PR (%)
IV	69.1/89.7	**70.5/91.9**
OPR	65.9/87.1	**67.8/89.9**
SV	68.1/87.0	**70.1/90.0**
OCC	65.2/84.5	**67.0/87.3**
DEF	67.2/90.5	**68.3/92.1**
MB	68.9/85.4	**71.6/89.1**
FM	66.8/84.5	**68.9/87.4**
IPR	67.6/89.5	**69.0/91.6**
OV	61.1/80.4	**63.9/84.0**
BC	63.0/81.1	**67.3/87.5**
LR	52.5/71.3	**62.3/84.9**
All	68.1/88.0	**69.6/90.3**

Generality analysis. Our method can be generalized to most of tracking frameworks, such as correlation filter, Siamese network, multi-domain network and transformer-based tracker. To show the generalization ability of the proposed method, we adopt three additional popular trackers with various frameworks, including RTMDNet[10], SiamRPN[24] and TransT[64]. Specifically, the search region is provided by the proposed motion-guided

modules, which predict the camera and object motion. Then, the trackers search the target in the corrected search region. We conduct analysis on OTB100 dataset. All the results in Table 6 show that trackers with our motion-guided module can achieve better performance.

Table 5 Ablation analysis of our method on OTB100 and MD200. Both motion modules make positive contributions to the final results with reasonable time cost. Thanks to the motion verifiers, reliable motion cues are extracted to assist the appearance tracker, thereby achieving better results. Our final model is superior to the baseline and maintains its speed.

Trackers	OTB100 SR/PR (%)	MD200 SR/PR (%)	Speed (fps)
DIMP	68.1/88.0	58.8/68.5	48.0
DIMP+CMD	66.7/86.4	59.0/69.1	39.0
DIMP+CMD+CMV	69.3/89.9	60.0/70.0	32.3
DIMP+CMM+OTD	57.2/73.0	48.0/54.2	29.8
DIMP+CMM+OTD+OTV	69.6/90.3	62.5/73.6	25.7

Table 6 Generality analysis for our motion-guided module on OTB100 dataset. Three additional popular frameworks are utilized to show the effectiveness of the proposed motion-guided module.

	RTMD Net	MG-RTMD Net	SiamRPN	MG-SiamRPN	TransT	MG-TransT
SR (%)	61.7	**63.4**	66.6	**67.0**	69.4	**70.1**
PR (%)	84.5	**86.7**	87.6	**88.0**	89.9	**90.3**

Analysis of the impact of loss in CMV. We analyze the influence of our designed loss for CMV. We test three versions of DIMP+CMM on OTB100. Table 7 presents the results. Each type of loss makes a positive contribution to the final performance.

Table 7 Impact of various losses on OTB100. Each loss makes a positive contribution to the final result.

Trackers	$\mathcal{L}_{cls}$	$\mathcal{L}_{con}$	$\mathcal{L}_{tri}$	SR (%)	PR (%)
DIMP+CMM	✓			67.6	87.7
DIMP+CMM	✓	✓		68.0	88.4
DIMP+CMM	✓	✓	✓	**68.5**	**89.0**

4.4 Qualitative analysis

Effectiveness of motion verifiers. We argue that our motion verifiers, i.e., the proposed CMV and OTV, can measure the quality of motion detectors. As shown in Fig. 6, the scores S_{OM} and S_{CM} are used to consider whether the motion results are reliable or not. Furthermore, we report the motion results with the search region provided by CMD and OTD, which are denoted as CMM SR, OTP SR, respectively. Due to camera motion and object motion, the target is at the boundary of the raw SR without our motion-guided module, our module can provide a prior for target location via motion detectors and select robust results to correct the search region using our verifier. Notably, our motion verifiers can provide accurate confidence, which is robust to appearance interference, including motion blur and illumination variation.

Qualitative comparison with baseline trackers. In this section, we depict qualitative results between DIMP and MG-DIMP in Fig. 7. DIMP with the proposed motion module can capture the object accurately, especially in fast motion, occlusion, and illumination. In sequence "deer", the target jumps continuously and motion blur occurs. The baseline tracker is confused with distinguishing the target against the similar distractors, therefore leading to drift. Our motion-guided module utilizes motion cues to supplement the appearance model effectively and can achieve a more robust tracker. In sequence "Girl2", the girl is occluded by the pedestrian, and DIMP fails to relocate the target when the girl reappears, while MG-DIMP succeeds in finding the target by predicting its trajectory. In sequence "Skating1", the target appearance changes dramatically due to the illumination, DIMP mistakes the surroundings as the target while motion cues can help track the target when appearance is not reliable for tracking. Furthermore, we provide some cases in MD200 to validate that our motion-guided module can boost the tracking accuracy and build a robust tracker, shown in Fig. 8. We state that our method shows great potential in handling large camera movement and partial occlusion. All the results highlight the importance of motion embedding and validate the effectiveness of our motion-guided module.

Parameter robustness analysis. We discuss the parameter robustness of our method. In the motion embedding process, reliable motion is selected by the step function, which contains two hyper-parameters. Fig. 9 shows that our method is robust to parameter disturbance and can perform well in a wide range of applications.

5 Speed analysis

In this section, we report the speed of different variants in Table 5. We note that each submodule is efficient for motion detection. Our final tracker (DIMP+CMM+OTD+OTV), with real-time speed, can achieve a more robust performance by combining both appearance and motion cues.

6 Conclusions

In this study, we emphasize the potential of motion cues and design a general and effective motion-guided module to fully exploit motion cues and guide appear-

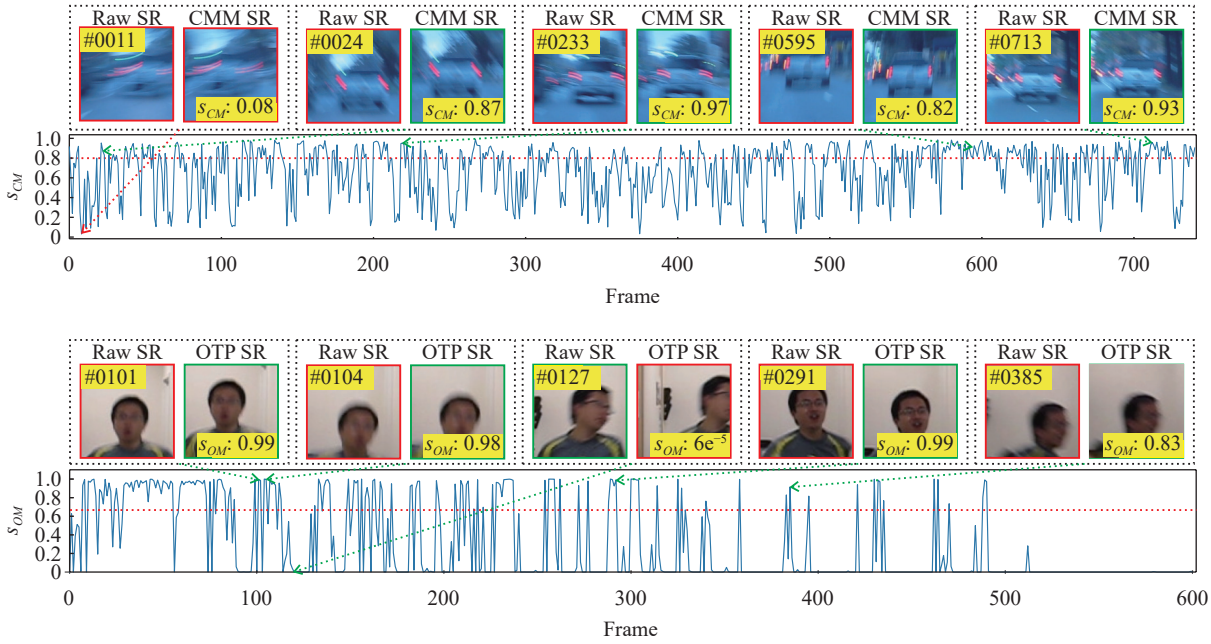


Fig. 6 Analysis of our proposed verifiers. We depict the original search region (Raw SR) and the search region corrected by CMM and OTP. Tracker with our motion verifier can distinguish whether the motion cue is reliable and robust to inference. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

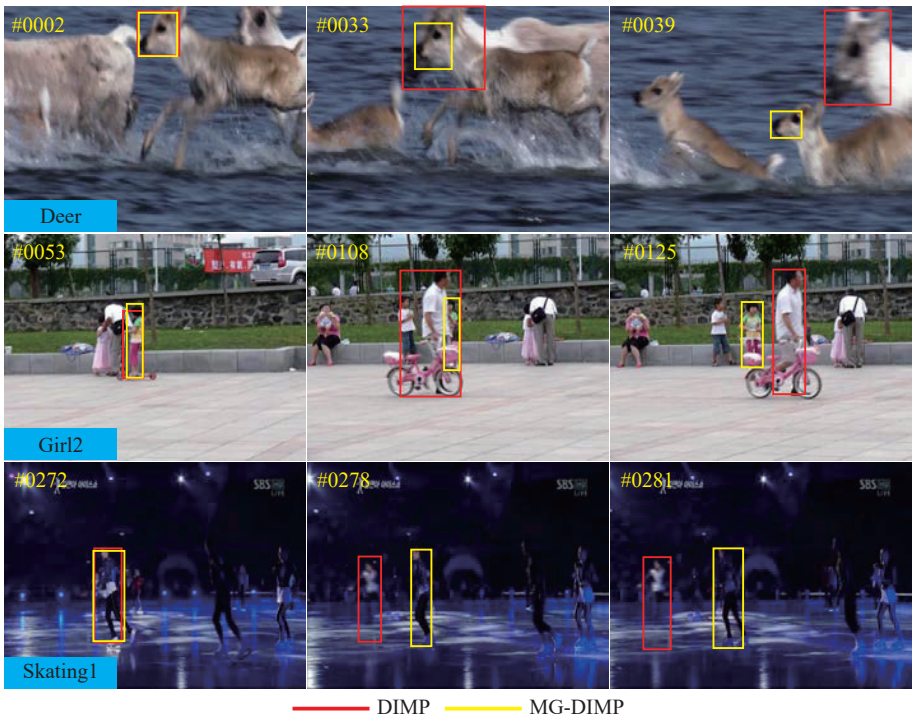


Fig. 7 Qualitative comparison between DIMP and MG-DIMP. The tracker with our motion-guided module is capable of fast motion, occlusion, and illumination variation. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

ance-based models. The motion cues are separated as camera motion and object motion. Two corresponding components are designed to handle them individually, namely camera motion modeling and object trajectory prediction. Each component consists of a motion detector and a verifier. As for camera motion modeling, we utilize off-the-shelf keypoint matching to model camera motion and evaluate the confidence of the detector via the proposed verifier, which is learned with numerous unlabeled data in a self-supervised learning paradigm. As for object trajectory prediction, we adopt a Conv-LSTM as our motion detector to handle sequential information and

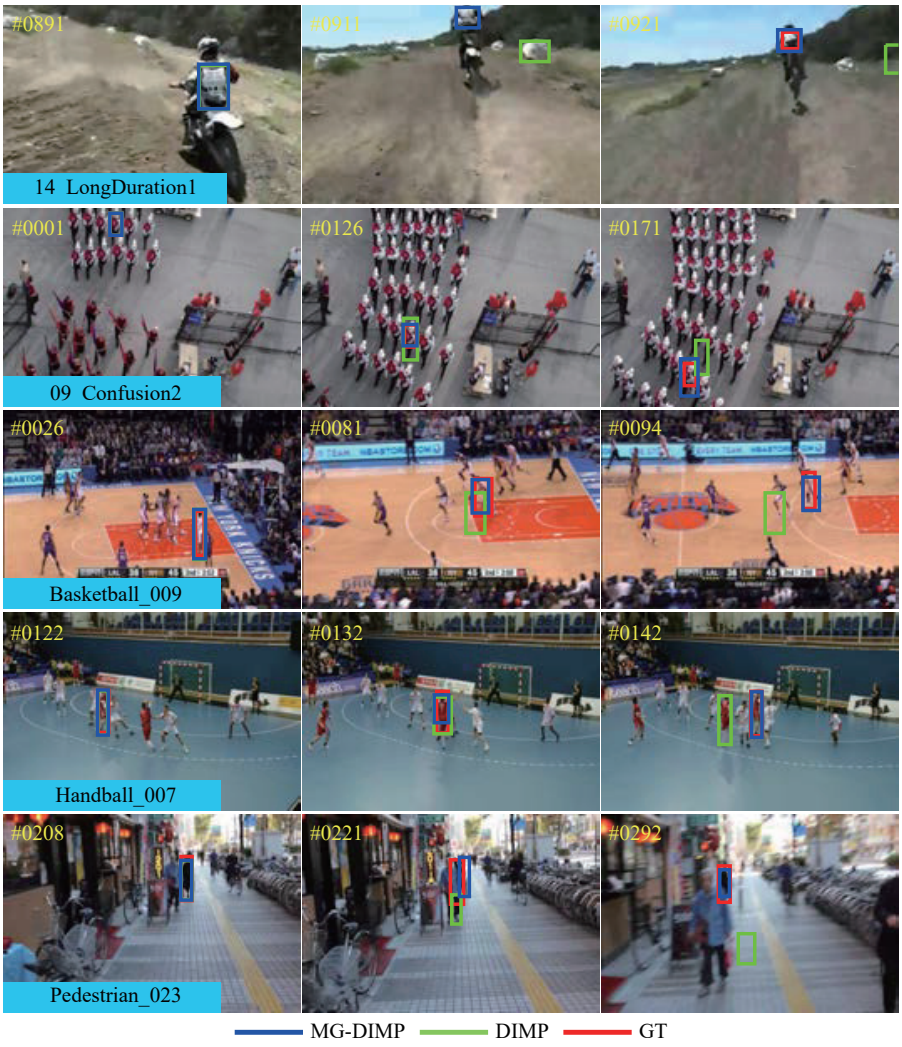


Fig. 8 Qualitative comparison on MD200. Compared with DIMP, MG-DIMP achieves substantial results in challenging cases, such as occlusion, fast motion, etc. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

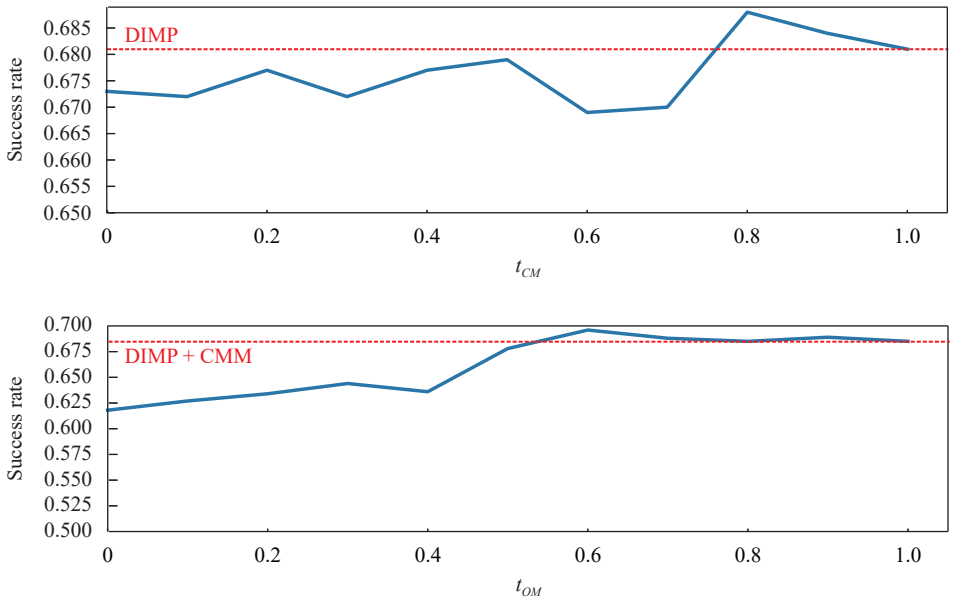


Fig. 9 Robustness analysis of our method. Both modules are parameter-insensitive and robust to parameter disturbance.

the results are measured by the proposed object trajectory verifier, which determines whether the predicted result is the target or not. Furthermore, we reconstruct a motion dataset, namely MD200, using the public tracking dataset to emphasize the importance of motion information. Numerous results on six popular datasets and MD200 demonstrate the effectiveness and generality of our method with a reasonable time cost.

Acknowledgements

The paper was supported in part by National Natural Science Foundation of China (Nos. 62293542 and 62022021), in part by Central Guidance on Local Science and Technology Development Fund of Liaoning Province, China (No. 2022JH6/100100026), in part by the National Defense Basic Scientific Research Program (No. WDZC202 15250205), and in part by Fundamental Research Funds for the Central Universities, China (No. DUT22QN228).

Declarations of conflict of interest

The authors declared that they have no conflicts of interest to this work.

References

[1] G. Welch, G. Bishop. An Introduction to the Kalman Filter, Technical Report TR 95-041, Department of Computer Science, University of North Carolina at Chapel Hill, USA, 2002.

[2] S. K. Weng, C. M. Kuo, S. K. Tu. Video object tracking using adaptive Kalman filter. *Journal of Visual Communication and Image Representation*, vol. 17, no. 6, pp. 1190–1208, 2006. DOI: 10.1016/j.jvcir.2006.03.004.

[3] J. Kwon, H. S. Lee, F. C. Park, K. M. Lee. A geometric particle filter for template-based visual tracking. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 4, pp. 625–643, 2014. DOI: 10.1109/TPAMI.2013.170.

[4] Y. Li, H. Ai, T. Yamashita, S. Lao, M. Kawade. Tracking in low frame rate video: A cascade particle filter with discriminative observers of different life spans. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1728–1740, 2008. DOI: 10.1109/TPAMI.2008.73.

[5] D. S. Bolme, J. R. Beveridge, B. A. Draper, Y. M. Lui. Visual object tracking using adaptive correlation filters. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, San Francisco, USA, pp. 2544–2550, 2010. DOI: 10.1109/CVPR.2010.5539960.

[6] M. Danelljan, G. Bhat, F. S. Khan, M. Felsberg. ECO: Efficient convolution operators for tracking. In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition*, Honolulu, USA, pp. 6931–6939, 2017. DOI: 10.1109/CVPR.2017.733.

[7] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, P. H. S. Torr. Fully-convolutional Siamese networks for object tracking. In *Proceedings of European Conference on Computer Vision*, Amsterdam, The Netherlands, pp. 850–865, 2016. DOI: 10.1007/978-3-319-48881-3_56.

[8] Q. Wang, L. Zhang, L. Bertinetto, W. Hu, P. H. S. Torr. Fast online object tracking and segmentation: A unifying approach. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 1328–1338, 2019. DOI: 10.1109/CVPR.2019.00142.

[9] H. Nam, B. Han. Learning multi-domain convolutional neural networks for visual tracking. In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, USA, pp. 4293–4302, 2016. DOI: 10.1109/CVPR.2016.465.

[10] I. Jung, J. Son, M. Baek, B. Han. Real-time MDNet. In *Proceedings of the 15th European Conference on Computer Vision*, Munich, Germany, pp. 89–104, 2018. DOI: 10.1007/978-3-030-01225-0_6.

[11] G. Bhat, M. Danelljan, L. Van Gool, R. Timofte. Learning discriminative model prediction for tracking. In *Proceedings of IEEE/CVF International Conference on Computer Vision*, Seoul, Republic of Korea, pp. 6182–6191, 2019. DOI: 10.1109/ICCV.2019.00628.

[12] S. Wei, L. Zhang, H. Liu. Integrated tracking and ISAR imaging using an integrated Kalman filter with wideband radar. *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 5, pp. 4639–4655, 2022. DOI: 10.1109/TAES.2022.3165006.

[13] W. Zhang, X. Zhao, Z. Liu, K. Liu, B. Chen. Converted state equation Kalman filter for nonlinear maneuvering target tracking. *Signal Processing*, vol. 202, Article number 108741, 2023. DOI: 10.1016/j.sigpro.2022.108741.

[14] G. Guo, S. Zhao. 3D multi-object tracking with adaptive cubature kalman filter for autonomous driving. *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 1, pp. 512–519, 2023. DOI: 10.1109/TIV.2022.3158419.

[15] D. Kim, Y. Han, H. Lee, Y. Kim, H. H. Kwon, C. Kim, W. Choi. Accelerated particle filter with GPU for real-time ballistic target tracking. *IEEE Access*, vol. 11, pp. 12139–12149, 2023. DOI: 10.1109/ACCESS.2023.3238873.

[16] M. K. Yılmaz, H. Bayram. Particle filter-based aerial tracking for moving targets. *Journal of Field Robotics*, vol. 40, no. 2, pp. 368–392, 2023. DOI: 10.1002/rob.22134.

[17] E. A. Olson, J. Pavlasek, J. A. Berry, O. C. Jenkins. Counter-hypothetical particle filters for single object pose tracking. In *Proceedings of IEEE International Conference on Robotics and Automation*, London, UK, pp. 3853–3859, 2023. DOI: 10.1109/ICRA48891.2023.10160625.

[18] J. Shin, S. Kim, S. Kang, S. W. Lee, J. Paik, B. Abidi, M. Abidi. Optical flow-based real-time object tracking using non-prior training active feature model. *Real-Time Imaging*, vol. 11, no. 3, pp. 204–218, 2005. DOI: 10.1016/j.rti.2005.03.006.

[19] J. F. Cohn, A. J. Zlochower, J. J. Lien, T. Kanade. Fast online object tracking and segmentation: A unifying approach. In *Proceedings of International Conference on Automatic Face and Gesture Recognition*, pp. 1328–1338, 1998.

[20] J. Šerých, J. Matas. Planar object tracking via weighted optical flow. In *Proceedings of IEEE/CVF Winter Confer-*

ence on Applications of Computer Vision, Waikoloa, USA, pp. 1593–1602, 2023. DOI: 10.1109/WACV56688.2023.00164.

[21] J. F. Henriques, R. Caseiro, P. Martins, J. Batista. High-speed tracking with kernelized correlation filters. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 37, no. 3, pp. 583–596, 2015. DOI: 10.1109/TPAMI.2014.2345390.

[22] D. Yuan, X. Chang, Z. Li, Z. He. Learning adaptive spatial-temporal context-aware correlation filters for UAV tracking. *ACM Transactions on Multimedia Computing Communications, and Applications*, vol. 18, no. 3, Article number 70, 2022. DOI: 10.1145/3486678.

[23] D. Yuan, X. Chang, Q. Liu, Y. Yang, D. Wang, M. Shu, Z. He, G. Shi. Active learning for deep visual tracking. *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 10, pp. 13284–13296, 2024. DOI: 10.1109/TNNLS.2023.3266837.

[24] B. Li, J. Yan, W. Wu, Z. Zhu, X. Hu. High performance visual tracking with siamese region proposal network. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Salt Lake City, USA, pp. 8971–8980, 2018. DOI: 10.1109/CVPR.2018.00935.

[25] S. Li, D. Y. Yeung. Visual object tracking for unmanned aerial vehicles: A benchmark and new motion models. In *Proceedings of the 31st AAAI Conference on Artificial Intelligence*, San Francisco, USA, pp. 4140–4146, 2017. DOI: 10.1609/aaai.v31i1.11205.

[26] Y. Liu, R. Li, Y. Cheng, R. T. Tan, X. Sui. Object tracking using spatio-temporal networks for future prediction location. In *Proceedings of the 16th European Conference on Computer Vision*, Glasgow, UK, 2020. DOI: 10.1007/978-3-030-58542-6_1.

[27] R. Hadsell, S. Chopra, Y. LeCun. Dimensionality reduction by learning an invariant mapping. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, New York, USA, pp. 1735–1742, 2006. DOI: 10.1109/CVPR.2006.100.

[28] E. Jang, C. Devin, V. Vanhoucke, S. Levine. Grasp2Vec: Learning object representations from self-supervised grasping. In *Proceedings of the 2nd Conference on Robot Learning*, Zürich, Switzerland, pp. 99–112, 2018.

[29] L. Tao, X. Wang, T. Yamasaki. Self-supervised video representation learning using inter-intra contrastive framework. In *Proceedings of the 28th ACM International Conference on Multimedia*, Seattle, USA, pp. 2193–2201, 2020. DOI: 10.1145/3394171.3413694.

[30] O. J. Henaff, A. Srinivas, J. De Fauw, A. Razavi, C. Doersch, S. M. A. Eslami, A. Van Den Oord. Data-efficient image recognition with contrastive predictive coding. In *Proceedings of the 37th International Conference on Machine Learning*, Article number 391, 2020.

[31] X. Zhai, A. Oliver, A. Kolesnikov, L. Beyer. S4L: Self-supervised semi-supervised learning. In *Proceedings of IEEE/CVF International Conference on Computer Vision*, Seoul, Republic of Korea, pp. 1476–1485, 2019. DOI: 10.1109/ICCV.2019.00156.

[32] R. Zhang, P. Isola, A. A. Efros. Colorful image colorization. In *Proceedings of the 14th European Conference on Computer Vision*, Amsterdam, The Netherlands, pp. 649–666, 2016. DOI: 10.1007/978-3-319-46487-9_40.

[33] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, A. A. Efros. Context encoders: Feature learning by inpainting. In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, USA, pp. 2536–2544, 2016. DOI: 10.1109/CVPR.2016.278.

[34] P. Agrawal, J. Carreira, J. Malik. Learning to see by moving. In *Proceedings of IEEE International Conference on Computer Vision*, Santiago, Chile, pp. 37–45, 2015. DOI: 10.1109/ICCV.2015.13.

[35] A. Mnih, K. Kavukcuoglu. Learning word embeddings efficiently with noise-contrastive estimation. In *Proceedings of the 27th International Conference on Neural Information Processing Systems*, Lake Tahoe, USA, pp. 2265–2273, 2013.

[36] F. Schroff, D. Kalenichenko, J. Philbin. FaceNet: A unified embedding for face recognition and clustering. In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition*, Boston, USA, pp. 815–823, 2015. DOI: 10.1109/CVPR.2015.7298682.

[37] T. Chen, S. Kornblith, M. Norouzi, G. Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning*, Article number 149, 2020.

[38] P. Zhang, W. Ouyang, P. Zhang, J. Xue, N. Zheng. SR-LSTM: State refinement for LSTM towards pedestrian trajectory prediction. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 12077–12086, 2019. DOI: 10.1109/CVPR.2019.01236.

[39] A. Mohamed, K. Qian, M. Elhoseiny, C. Claudel. Social-STGCNN: A social spatio-temporal graph convolutional neural network for human trajectory prediction. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Seattle, USA, pp. 14412–14420, 2020. DOI: 10.1109/CVPR42600.2020.01443.

[40] J. Liang, L. Jiang, K. Murphy, T. Yu, A. Hauptmann. The garden of forking paths: Towards multi-future trajectory prediction. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Seattle, USA, pp. 10505–10515, 2020. DOI: 10.1109/CVPR42600.2020.01052.

[41] Y. C. Tang, R. Salakhutdinov. Multiple futures prediction. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, Vancouver, Canada, Article number 1382, 2019.

[42] R. Chandra, U. Bhattacharya, A. Bera, D. Manocha. TraPHic: Trajectory prediction in dense and heterogeneous traffic using weighted interactions. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 8475–8484, 2019. DOI: 10.1109/CVPR.2019.00868.

[43] N. Mohajerin, M. Rohani. Multi-step prediction of occupancy grid maps with recurrent neural networks. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 10592–10600, 2019. DOI: 10.1109/CVPR.2019.01085.

[44] P. Zhang, J. Zhao, C. Bo, D. Wang, H. Lu, X. Yang. Jointly modeling motion and appearance cues for robust RGB-T tracking. *IEEE Transactions on Image Processing*,

vol. 30, pp. 3335–3347, 2021. DOI: 10.1109/TIP.2021.3060862.

[45] H. Bay, T. Tuytelaars, L. Van Gool. SURF: Speeded up robust features. In *Proceedings of the 9th European Conference on Computer Vision*, Graz, Austria, pp. 404–417, 2006. DOI: 10.1007/11744023_32.

[46] D. G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004. DOI: 10.1023/B:VISI.0000029664.99615.94.

[47] Y. Ono, E. Trulls, P. Fua, K. M. Yi. LF-Net: Learning local features from images. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, Montreal, Canada, pp. 6237–6247, 2018.

[48] J. Revaud, P. Weinzaepfel, C. De Souza, M. Humenberger. R2D2: Repeatable and reliable detector and descriptor. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, Vancouver, Canada, Article number 1113, 2019.

[49] B. Jiang, R. Luo, J. Mao, T. Xiao, Y. Jiang. Acquisition of localization confidence for accurate object detection. In *Proceedings of the 15th European Conference on Computer Vision*, Munich, Germany, pp. 816–832, 2018. DOI: 10.1007/978-3-030-01264-9_48.

[50] H. Fan, L. Lin, F. Yang, P. Chu, G. Deng, S. Yu, H. Bai, Y. Xu, C. Liao, H. Ling. LaSOT: A high-quality benchmark for large-scale single object tracking. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 5369–5378, 2019. DOI: 10.1109/CVPR.2019.00552.

[51] D. P. Kingma, J. Ba. Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations*, San Diego, USA, 2015.

[52] Y. Wu, J. Lim, M. H. Yang. Object tracking benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 37, no. 9, pp. 1834–1848, 2015. DOI: 10.1109/TPAMI.2014.2388226.

[53] P. Liang, E. Blasch, H. Ling. Encoding color information for visual tracking: Algorithms and benchmark. *IEEE Transactions on Image Processing*, vol. 24, no. 12, pp. 5630–5644, 2015. DOI: 10.1109/TIP.2015.2482905.

[54] M. Mueller, N. Smith, B. Ghanem. A benchmark and simulator for UAV tracking. In *Proceedings of the 14th European Conference on Computer Vision*, Amsterdam, The Netherlands, pp. 445–461, 2016. DOI: 10.1007/978-3-319-46448-0_27.

[55] M. Kristan, A. Leonardis, J. Matas, M. Felsberg, R. Pflugfelder, L. Č . Zajc, T. Vojíř, G. Bhat, A. Lukežič, A. Eldesokey, G. Fernández, Á . García-Martín, Á . Iglesias-Arias, A. A. Alatan, A. González-García, A. Petrosino, A. Memarmoghadam, A. Vedaldi, A. Muhič, A. He, A. Smeulders, A. G. Perera, B. Li, B. Chen, C. Kim, C. Xu, C. Xiong, C. Tian, C. Luo, C. Sun, C. Hao, D. Kim, D. Mishra, D. Chen, D. Wang, D. Wee, E. Gavves, E. Gundogdu, E. Velasco-Salido, F. S. Khan, F. Yang, F. Zhao, F. Li, F. Battistone, G. De Ath, G. R. K. S. Subrahmanyam, G. Bastos, H. Ling, H. K. Galoogahi, H. Lee, H. Li, H. Zhao, H. Fan, H. Zhang, H. Possegger, H. Li, H. Lu, H. Zhi, H. Li, H. Lee, H. J. Chang, I. Drummond, J. Valmadre, J. S. Martin, J. Chahl, J. Y. Choi, J. Li, J. Wang, J. Qi, J. Sung, J. Johnander, J. Henriques, J. Choi, J. van de Weijer, J. R. Herranz, J. M. Martínez, J. Kittler, J. Zhuang, J. Gao, K. Grm, L. Zhang, L. Wang, L. Yang, L. Rout, L. Si, L. Bertinetto, L. Chu, M. Che, M. E. Maresca, M. Danelljan, M. H. Yang, M. Abdelpakey, M. Shehata, M. Kang, N. Lee, N. Wang, O. Miksik, P. Moallem, P. Vicente-Moñivar, P. Senna, P. Li, P. Torr, P. M. Raju, R. Qian, Q. Wang, Q. Zhou, Q. Guo, R. Martín-Nieto, R. K. Gorthi, R. Tao, R. Bowden, R. Everson, R. Wang, S. Yun, S. Choi, S. Vivas, S. Bai, S. Huang, S. Wu, S. Hadfield, S. Wang, S. Golodetz, T. Ming, T. Xu, T. Zhang, T. Fischer, V. Santopietro, V. Štruc, W. Wei, W. Zuo, W. Feng, W. Wu, W. Zou, W. Hu, W. Zhou, W. Zeng, X. Zhang, X. Wu, X. J. Wu, X. Tian, Y. Li, Y. Lu, Y. W. Law, Y. Wu, Y. Demiris, Y. Yang, Y. Jiao, Y. Li, Y. Zhang, Y. Sun, Z. Zhang, Z. Zhu, Z. H. Feng, Z. Wang, Z. He. The sixth visual object tracking VOT2018 challenge results. In *Proceedings of European Conference on Computer Vision*, Munich, Germany, pp. 3–53, 2018. DOI: 10.1007/978-3-030-11009-3_1.

[56] M. Kristan, J. Matas, A. Leonardis, T. Vojir, R. Pflugfelder, G. Fernandez, G. Nebehay, F. Porikli, L. Cehovin. A novel performance evaluation methodology for single-target trackers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 11, pp. 2137–2155, 2016. DOI: 10.1109/TPAMI.2016.2516982.

[57] L. Huang, X. Zhao, K. Huang. GOT-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 5, pp. 1562–1577, 2021. DOI: 10.1109/TPAMI.2019.2957464.

[58] A. W. M. Smeulders, D. M. Chu, R. Cucchiara, S. Calderara, A. Dehghan, M. Shah. Visual tracking: An experimental survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 7, pp. 1442–1468, 2014. DOI: 10.1109/TPAMI.2013.230.

[59] H. K. Galoogahi, A. Fagg, C. Huang, D. Ramanan, S. Lucey. Need for speed: A benchmark for higher frame rate object tracking. In *Proceedings of IEEE International Conference on Computer Vision*, Venice, Italy, pp. 1134–1143, 2017. DOI: 10.1109/ICCV.2017.128.

[60] A. Li, M. Lin, Y. Wu, M. H. Yang, S. Yan. NUS-PRO: A new visual tracking challenge. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 2, pp. 335–349, 2016. DOI: 10.1109/TPAMI.2015.2417577.

[61] T. Xu, Z. H. Feng, X. J. Wu, J. Kittler. Learning adaptive discriminative correlation filters via temporal consistency preserving spatial feature selection for robust visual object tracking. *IEEE Transactions on Image Processing*, vol. 28, no. 11, pp. 5596–5609, 2019. DOI: 10.1109/TIP.2019.2919201.

[62] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, J. Yan. SiamRPN++: Evolution of Siamese visual tracking with very deep networks. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, USA, pp. 4277–4286, 2019. DOI: 10.1109/CVPR.2019.00441.

[63] B. Liao, C. Wang, Y. Wang, Y. Wang, J. Yin. PG-Net: Pixel to global matching network for visual tracking. In *Proceedings of the 16th European Conference on Computer Vision*, Glasgow, UK, pp. 429–444, 2020. DOI: 10.1007/978-3-030-58542-6_26.

[64] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, H. Lu. Trans-

former tracking. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Nashville, USA, pp. 8122–8131, 2021. DOI: 10.1109/CVPR46437.2021.00803.



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Citation: P. Zhang, S. Lai, D. Wang, H. Lu. Motion–guided visual tracking. *Machine Intelligence Research*, vol.22, no.5, pp.983–998, 2025. https://doi.org/10.1007/s11633–023–1477–x

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